

UNIVERSAL INSULATED FLAT ROOF SYSTEM

WARM FLAT ROOF WITH PIR INSULATION AND FLEXIBLE WATERPROOFING OPTIONS

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SYSTEM OVERVIEW

The universal warm flat roof combines a structural timber deck, continuous vapour control, aluminium-faced PIR insulation and a durable waterproofing system. Different waterproofing materials can be selected according to roof geometry, exposure, detailing and project requirements.

Final design must verify structure, falls, drainage, wind uplift, vapour control, insulation, waterproofing compatibility, penetrations, perimeters and fire requirements.

WHY ECOVIVA

- Independent renovation guidance
- Local Mallorca expertise
- Project-specific technical verification
- One point of contact from concept to completion

SIX-PART WARM FLAT ROOF BUILD-UP

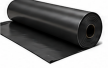
- Structural timber joists**
Primary roof structure sized for the project spans and loads.
- Structural OSB/3 deck**
Rigid load-bearing deck providing a stable roof substrate.
- V3 vapour barrier**
Continuous vapour-control layer below the insulation.
- Aluminium-faced PIR insulation**
High-performance rigid insulation forming a continuous thermal layer.
- Waterproofing system**
Selected membrane forming the durable weather-exposed roof covering.
- Drainage and perimeter detailing**
Falls, drains, flashings and edge profiles direct water safely away.

KEY TECHNICAL PRINCIPLES

- CONTINUOUS VAPOUR CONTROL**
Seal the vapour barrier at laps, penetrations and perimeter connections.
- DESIGNED ROOF FALLS**
Provide sufficient falls toward outlets without persistent ponding.
- SECURE PERIMETERS AND PENETRATIONS**
Detail edges, upstands, drains, rooflights and services carefully.
- CONTINUOUS PIR INSULATION**
Keep the insulation layer uninterrupted to reduce thermal bridging.
- COMPATIBLE WATERPROOFING**
Use primers, membranes, adhesives and accessories as one system.
- WIND, FIRE AND MOISTURE VERIFICATION**
Verify wind uplift, fire performance, condensation and fixings.

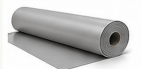
EXAMPLES OF COMPONENTS USED IN THE SYSTEM

Components are selected for the deck, insulation, waterproofing, geometry, exposure and project requirements.

							
Roof waterproofing system Selected membrane or liquid finish.	V3 black vapour barrier Continuous vapour-control layer.	Aluminium-faced PIR insulation High-performance rigid insulation.	Roof drain with leaf guard Rainwater outlet with debris guard.	Aluminium roof edge profile Edge trim with fascia and drip.	Zinc / galvanised flashing Formed metal for junctions and upstands.	Stainless-steel fixings Corrosion-resistant roof fixings.	Structural timber joist Primary member sized for roof loads.

ROOF WATERPROOFING OPTIONS

Select the waterproofing for the geometry, substrate, exposure, detailing, maintenance and project requirements.

	BITUMINOUS ROOFING Two-layer torch-on system with robust, established detailing. REFERENCE SYSTEM SHOWN IN HERO		RESITRIX® EPDM HYBRID MEMBRANE Reinforced EPDM with bituminous backing and welded joints.		EPDM SINGLE-PLY MEMBRANE Flexible rubber membrane with strong weather and UV resistance.
	PVC ROOFING MEMBRANE Lightweight thermoplastic membrane with welded seams.		TPO ROOFING MEMBRANE Recyclable thermoplastic membrane resistant to heat and UV.		LIQUID WATERPROOFING SYSTEM Seamless PMMA or PU solution for complex details and repairs.

KEY BENEFITS

- High thermal performance
- Continuous external roof insulation
- Reduced thermal bridging
- Flexible waterproofing selection
- Reliable rainwater drainage
- Durable weather protection
- Adaptable perimeter detailing
- Renovation and new-build use

TYPICAL APPLICATIONS

- Timber flat roofs
- Villa roof renovations
- Residential extensions
- Garages and annexes
- Apartment and commercial buildings
- New-build warm flat roofs

WATERPROOFING SELECTION AND SERVICE LIFE

Performance depends on system choice, preparation, detailing, drainage, exposure and maintenance. Compatibility, joints, reparability and roof access must be reviewed.

IMPORTANT TECHNICAL REQUIREMENTS

Verify joist sizing; deck condition and structural capacity; Provide continuous falls to correctly sized outlets; Seal vapour-control laps, penetrations and perimeters; Maintain continuous aluminium-faced PIR insulation; Use compatible primers, membranes and accessories; Secure perimeters and corners for wind uplift; Detail drains, upstands, rooflights and penetrations; Provide sufficient waterproofing upstand heights; Coordinate edge profiles, flashings and façades; Avoid trapped water and persistent ponding; Verify condensation and internal humidity; Confirm fire, maintenance and access requirements; Allow for Mallorca sun, coast and intense rainfall; Follow the CTE, product data and project requirements.